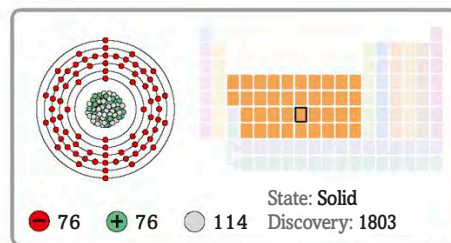
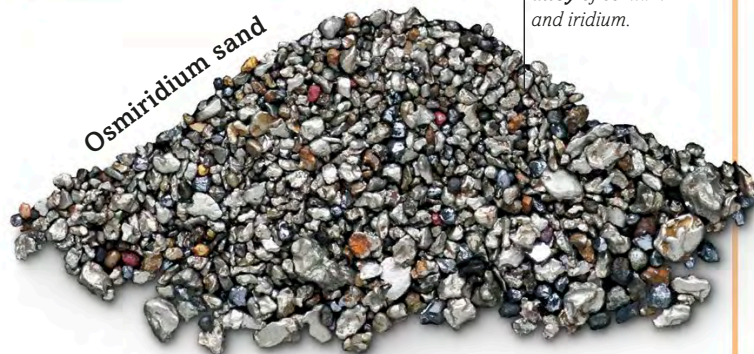


76
Os

Osmium



Forms



Osmiridium sand

This is a natural alloy of osmium and iridium.

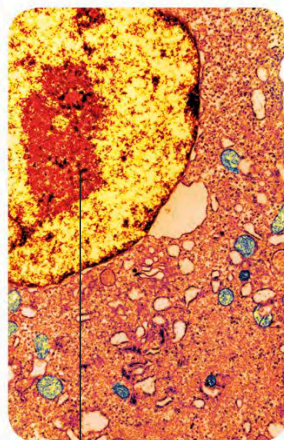
The pure form is hard but shatters easily.

Pellet of pure osmium refined in a laboratory



Uses

Transmission electron microscope (TEM) image



Osmium oxide is used to highlight objects inside a cell.

Fingerprint powder in use



Black osmium oxide powder clings to oily fingerprints.



Record player

The needle of this old record player is made of osmium.

Fountain pen



The nib of this pen moves smoothly because of its hardy osmium alloy.

Osmium is the densest of all naturally occurring elements: 250 ml (8.5 fl oz) of this metal (in its liquid form) weighs 5.5 kg (12 lb). This rare element is found in the ore **osmiridium**. **Pure osmium** reacts with oxygen in the air to form a poisonous oxide, so the metal is used safely

by combining it with other elements or alloys. A red osmium oxide stains cells so they can be seen clearly under a powerful **microscope**, while a black oxide powder allows **fingerprints** to be revealed in crime investigations. A hard osmium alloy is used in **fountain pen** nibs.